d. Since proteins are macro molecules, their sizes are quantitatively expressed in terms of their molecular weights with kilo Daltons as their unit (kDa). For example the human serum albumin has a molecular weight of 66 kDa.

e. Because of their giant size, proteins exhibit colloidal properties such as low diffusion and Tyndall effect.

f. Proteins undergo hydrolysis upon treatment with concentrated mineral acids like HCl and yield constituent amino acids as their hydrochlorides. Similarly, proteolytic enzymes like trypsin and chymotrypsin hydrolyse proteins.

g. When proteins are treated with alkaline copper sulphate solution (Biuret reagent) they form a violet coloured complex called Biuret complex. This reaction can be used as a qualitative and quantitative test for proteins.

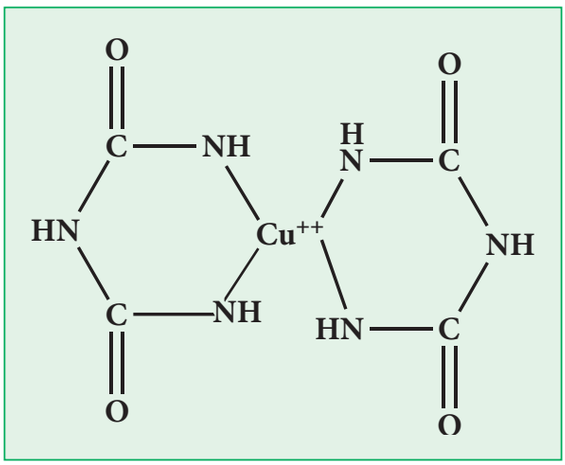

Figure 3.13 Biuret complex

3.5 Haemoglobin – an example for globular protein

Haemoglobin is found in red blood cells and is involved in the transport of oxygen from lungs to tissues. It is a tetramer containing four polypeptide chains – 2α chains and 2 β chains. Each of these chains contains a prosthetic group called heme. Heme is a protoporphyrin ring complexed with Fe^{2+}. This Fe^{2+} ion can form six bonds, four with the nitrogen atoms of the porphyrin ring, one with a histidine of hemoglobin and the other with oxygen. Thus every haemoglobin can carry four O_2 molecules.

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